

Name:	B/o Jeevitha	DOB: 18/10/2022	Sex:
Sampling date: 04/11/2022		Received on: 07/11/2022	
Ref Dr:	Dr. D V Suresh, Dr. Hemnath	File # : 1311/22	
Hosp/ City:	SKS Hospital, Salem		

Basic Urinary Studies – Metabolic Routine Screen Tests

A	Urine MRST (Metabolic Routine screen tests)		
	Smell: ND	Colour: Pale yellow	
	Urine pH: 6.0	Quantity: 14 ml	
1	Urine Creatinine	- mMol/L	
2	Ketostix test (Strip method)	Negative	
3	Glucostix Test (Strip method)	Negative	
4	Sulfites Test (Strip method)	Negative	Sulfite Oxidase def
5	C.T.A.B.	Negative	M.P.S.
6	Benedict's Test	Negative	Reducing Substances
7	Seilwinoff's Test	Negative	Fructosuria
8	D.N.P.H. test	Negative	Keto-acids, MSUD / PKU
9	Ferric Chloride test	Negative	Keto-acids Phosphates
10	Bratton Marshall Test	Negative	ADSL def.
B	TLC Sugar	NA	
C	TLC Oligosaccharides	NA	

Impression: Normal urine MRST. Sulfites are not detectable in the current sample.

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MD DCH MCPS

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Biotinidase Enzyme Activity

Date: 08/11/2022

Method: Wolf's colorimetric method.

Sample: Serum

Result	Serum: nmol / min / ml Normal : 4.5 – 7.5 Partial : 2.5 – 4.5 Deficient : Less than 2.5	Dried blood spots: OD Deficient: OD < 0.010 Partial : OD:0.010- 0.030 Normal: OD > 0.030	Impression
1 st Reading	6.23 nmol / min / ml	-	
2 nd Reading	6.31 nmol / min / ml	-	
Average	6.27 nmol / min / ml	-	NORMAL

Impression: Normal Biotinidase enzyme activity.

- Elevated Biotinidase enzyme activity may be seen in Glycogen Storage Disorders.
- Normal Biotinidase enzyme activity does not rule out Biotin deficiency due to other reasons like dietary deficiency, absorption defects [malabsorption syndrome], SMVT gene defects or high intake of raw eggs etc.
- Classical Severe Biotinidase enzyme deficiency should have activity less than 5 % of normal. Activities up to 10 % are also considered deficient.
- Partial deficiency is generally not considered to be symptomatic unless associated with some other stressful conditions like Infection , Carnitine Deficiency etc.
- Partial deficiency has also been noted in Organic Acidemias involving Carboxylases enzymes or other Organic Acidemias e.g. PA / IVA / MMA.
- Biotinidase deficiency is an autosomal recessively inherited disorder, which is now days considered as treatable by Tab. Biotin supplementation.
- In cases of partial deficiency it is always necessary to rule out other IEMs by MS/MS and GC/MS.
- Neonatal screening for this disorder has also been recommended routinely.

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Carnitine / Acyl Carnitine Profile – Derivatization method - DBS

Date: 08/11/2022

Sample: Dried Blood spots

Method: Derivatization method

Instrument: LC/MS/MS (Agilent LC6420)

	Carnitine / Acyl Carnitine		Normal Range (uMol/L)
1	Total Carnitine	41.70	20 – 87.7
2	Free Carnitine	16.54	24.7 – 66.6 (<8.6 for CTD)
3	Acyl Carnitine	25.16	
4	Free / Acyl ratio	0.66	> 2.0
5	Acyl Carnitine Profile	Normal acyl carnitine profile	
	Amino acids		Normal Range (uMol/L)
1	Alanine	541.21	650.0
2	Glycine	373.86	1000.0
3	Leucine - Isoleucine	42.75	300.0
4	Valine	38.66	300.0
5	Methionine	29.67	75.0
6	Phenylalanine	49.35	150.0
7	Tyrosine	49.82	400.0
8	Arginine	17.88	70.0
9	Citrulline	13.22	55.0
10	Ornithine	280.69	393.0
11	Glutamic acid	352.50	1070.0
12	Aspartic acid	35.65	737.0

Impression: Normal total carnitine, low free carnitine and low free / acyl carnitine ratio. Normal levels of amino acids.

Adv: Please correlate clinically.

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Carnitine / Acyl Carnitine Profile – Derivatization method - DBS

	Acyl Carnitines	Values	Cut-offs (uMol/L)	Ratios	Values	Cut-Offs
1	C 2	20.22	25.0	C3/C0		0.20
2	C 3	1.31	6.00	C3/C2		0.20
3	C 3DC	0.03	0.30	C3/C16		3.50
4	C 4	0.29	1.30	C4/C2		0.10
5	C 4-OH	0.10	0.65	C4/C3		0.70
6	C 4DC	0.02	0.63	C4OH/C2		0.026
7	C 5	0.24	0.70	C5/C2		0.05
8	C 5-OH	0.08	0.90	C5-OH/C2		0.04
9	C 5:1	0.02	0.25	C5:1/C2		0.20
10	C 5 DC	0.03	0.35	C5DC/C2		0.02
11	C 6	0.02	0.45	C5DC / C12		
12	C6DC	0.11	0.87			
13	C 8	0.03	0.45	C8/C2		0.03
14	C8:1	0.06	1.50			
15	C 10	0.04	0.45	C10:1/C2		
16	C 10: 1	0.04	0.30			
17	C 10: 2	0.01	0.15			
18	C 12	0.03	0.50			
19	C 14	0.08	0.80	C14/C2		0.03
20	C14OH	0.01	0.04			
21	C 14: 1	0.07	0.60	C14:1/C16		0.30
22	C14 : 2	0.02	0.15			
23	C 16	0.89	7.50			
24	C16:1	0.05	0.35			
25	C16: 1OH	0.03	0.14			
26	C 16-OH	0.01	0.15	C16-OH/C2		0.01
27	C 18	0.51	2.50	C18 :1/C2		0.15
28	C 18-OH	0.02	0.10			
29	C 18: 1	0.65	3.50	C18-OH/C2		0.10
30	C 18: 1-OH	0.01	0.16	C18:1OH/C2		0.01
31	C18:2	0.11	0.90			
32	C16DC	0.01	0.10			
33	C18:1DC	0.01	0.06			

Imp: Normal acyl carnitine profile.

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Note:

1. Metabolic lab reports should not be interpreted in isolation and should always be correlated with clinical findings and with other medical reports
2. The results are to be used as guidance by medical professionals only.
3. The results provided above are semi-quantitative as they are performed on dried blood spots and they relate only to the sample provided. In case of clinical suspicion of any specific disease, quantitative assays may be performed on liquid blood samples.
4. Additional studies including plasma amino acids, urine organic acids and quantitation of metabolites may be needed in case child develops any symptoms related to metabolic disorders.
5. Certain disorders may develop slowly or may have intermittent elevation of certain metabolites.
6. Some values may be altered by drugs received by the patient prior to sampling. This should be considered while interpreting the lab reports
7. Normal amino acids and acyl carnitine profile does not rule out all Inborn Errors of Metabolism
8. The above method is adapted from method described by Rinaldo P et al. (Rinaldo P, et al. Acylcarnitine profile analysis. ACMG standards and Guidelines. Genetics in Medicine, 2008. Vol 10 No 2.)
9. The above results may be published only after approval from the authorized signatory.

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AMINO ACIDS – Plasma – UHPLC

Date: 09/11/2022

Sr. No	Test	Plasma (μmol/L)	Reference range (μmol/L)
1	Aspartic Acid	87	34 – 94
2	Glutamic Acid	54	17 – 69
3	Asparagine	83	28 – 65
4	Serine	296	92 – 196
5	Glutamine	384	457 – 857
6	Histidine	124	68 – 108
7	Glycine	717	166 – 450
8	Threonine	88	102 – 246
9	Citrulline	14	19 – 52
10	Alanine	834	242 – 594
11	Arginine	55	1 – 81
12	Tyrosine	39	35 – 107
13	Cystine	24	36 – 58
14	Valine	92	155 – 343
15	Methionine	50	13 – 60
16	Tryptophan	20	10 – 140
17	Phenylalanine	56	34 – 86
18	Iso – Leucine	19	34 – 106
19	Ornithine	163	47 – 195
20	Leucine	127	86 – 206
21	Lysine	237	116 – 276
22	Proline	157	58 – 324
23	Total Amino acids	3720	3000 - 5000
24	Total BCAA	238	< 600
25	Gly / BCAA	3.01	
26	Ala / BCAA	3.51	
27	Gly / Total	20%	< 10 %
28	Ala/ Total	23%	< 10%

Imp: Mildly elevated glycine and alanine.

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AMINO ACIDS – CSF – UHPLC

Date: 09/11/2022

Sr. No	Test	Observed values ($\mu\text{mol/L}$)	Reference range
1	Aspartic Acid	17	6 – 18
2	Glutamic Acid	2	1 – 31
3	Asparagine	7	6 – 14
4	Serine	67	18 – 38
5	Glutamine	428	356 – 680
6	Histidine	44	11 – 25
7	Glycine	55	2 – 14
8	Threonine	25	22 – 53
9	Citrulline	3	1 – 5
10	Alanine	15	13 – 48
11	Arginine	3	13 – 35
12	Tyrosine	5	5 – 14
13	Cystine	2	-
14	Valine	7	10 – 38
15	Methionine	8	0 – 9
16	Tryptophan	1	
17	Phenylalanine	11	6 – 19
18	Iso – Leucine	4	3 – 13
19	Ornithine	5	3 – 9
20	Leucine	12	10 – 27
21	Lysine	8	15 – 43
22	Hydroxyproline		-
23	Proline	15	-
24	CSF / plasma glycine	0.07	0.02-0.08

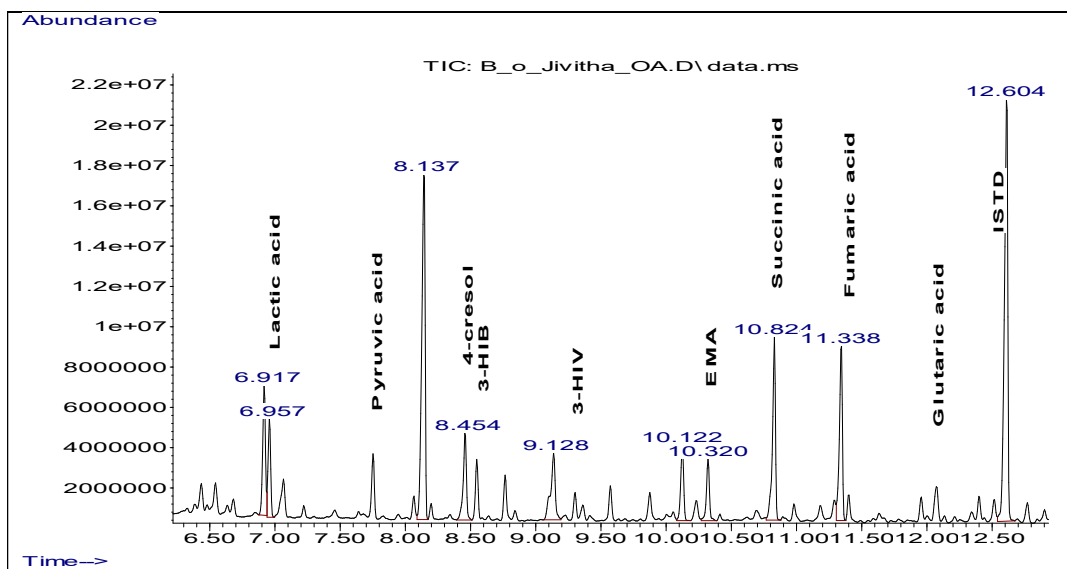
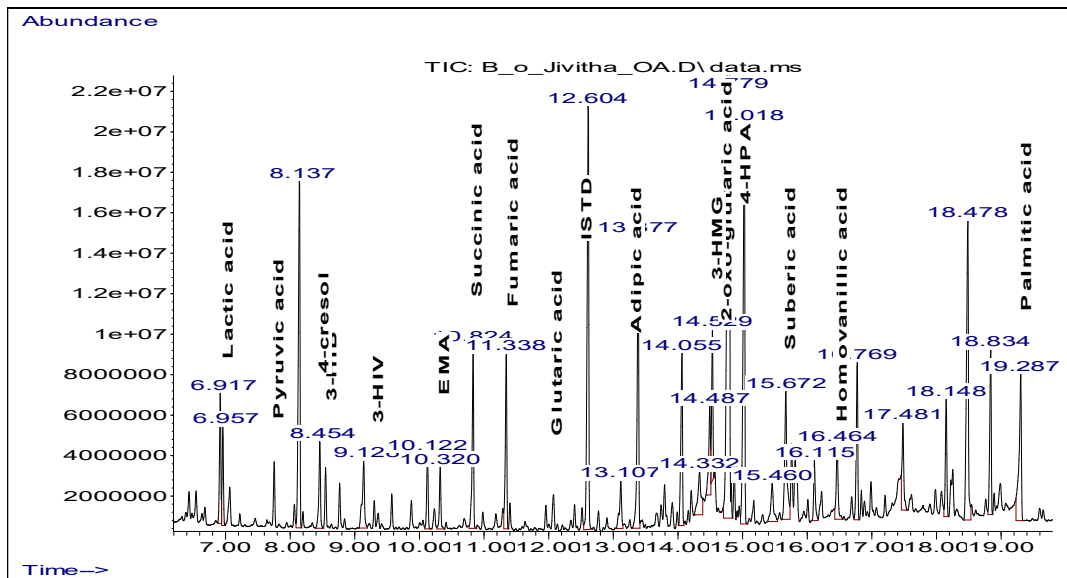
Imp: Mildly elevated glycine.

Adv: Please correlate clinically.

Dr. Anil B Jalan
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Pediatric Geneticist

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GC-MS report of Organic Acid Analysis- Reviewed Report



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GC-MS report of Organic Acid Analysis- Reviewed Report

RT	Compound
6.957	Lactic acid
7.750	Pyruvic acid
8.454	4-cresol
8.547	3-hydroxyisobutyric acid (3-HIB)
9.301	3-hydroxyisovaleric acid (3-HIV)
10.320	Ethylmalonic acid (EMA)
10.824	Succinic acid
11.338	Fumaric acid
12.071	Glutaric acid
12.604	ISTD
13.377	Adipic acid
14.529	3-hydroxy-3-methylglutaric acid
14.779	2-oxo-glutaric acid
15.018	4-hydroxyphenylacetic acid (4-HPA)
15.672	Suberic acid
16.464	Homovanillic acid
19.287	Palmitic acid

Interpretation: Normal urine organic acid profile. No sign of any specific organic or dicarboxylic aciduria.

Adv.: Please correlate with:

- 1) Carnitine / acyl carnitine profile in blood
- 2) Plasma amino acids

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Pipecolic Acid and AASA in CSF

Date: 10/11/2022

Parameter	Result	Reference Range
Pipecolic Acid	0.27 umol/L	0.01 – 1.00 umol / L
AASA	0.38 umol/L	0 – 2.00 umol/L

Impression: Normal levels of pipecolic acid and AASA in CSF.

Adv:

1. Please correlate clinically and with EEG.
2. **Please correlate with plasma and urinary pipecolic acid and AASA** as sometimes CSF pipecolic acid may not be elevated
3. If clinically pyridoxine dependent seizures are suspected then Antiquitin gene study is recommended.

Note: Pipecolic acid may be elevated with the following conditions:

- Antiquitin gene defect
- Peroxisomal disorders (Zellweger spectrum disorders, atypical Refsum disorder)
- Acute hepatocellular dysfunction
- Chronic liver disease
- Hyperlysinemia
- Defects in Proline Metabolism.
- Benign trait
- Joubert's syndrome
- Dyggve – Melchior Clausen Syndrome
- Hyperthyroidism
- Glutaric aciduria type II

AASA can be elevated in Molybdenum cofactor deficiency or sulfite oxidase deficiency

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MD DCH MCPS

BIOCHEMICAL GENETIC REPORT

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Hosp/ City:	SKS Hospital, Salem		

Pipecolic Acid and AASA in Urine

Date: 10/11/2022

Creatinine: 2.12 mmol/L

Parameter	Result uM/L	Result uM/mM creat	Reference Range
Pipecolic Acid	5.37 uM/ L	2.53 umol/ mmol creat	< 6 months: 0.55 – 24.1 > 6 months : 0.01 – 1.54
AASA	0.77 uM/ L	0.36 umol/ mmol creat	0 – 2.0 umol/ mmol creat

Impression: Normal levels of pipecolic acid and AASA in urine.

Adv: Please correlate with CSF or plasma pipecolic acid and CSF 5-MTHF levels.

Note: Pipecolic acid may be elevated with the following conditions:

- Antiquitin gene defect
- Peroxisomal disorders (Zellweger spectrum disorders, atypical Refsum disorder)
- Acute hepatocellular dysfunction
- Chronic liver disease
- Hyperlysinemia
- Defects in Proline Metabolism.
- Benign trait
- Joubert's syndrome
- Dyggve – Melchior Clausen Syndrome
- Hyperthyroidism
- Glutaric aciduria type II

AASA can be elevated in Molybdenum cofactor deficiency or sulfite oxidase deficiency

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Paediatric Geneticist

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Hosp/ City:	SKS Hospital, Salem		

Date: 11/11/2022

To
Dr. D V Suresh and Dr. Hemnath
SKS Hospital, Salem

Dear Sir

Many thanks for referring samples of B/o Jeevitha with history of poor feeding and seizures and EEG showing multiple epileptiform discharges. We have completed the metabolic studies as per your request and following are the results:

1. Urine Sulfites: Normal
2. Serum Biotinidase enzyme: Normal
3. TMS of blood: shows low free carnitine (16.54 umol/L) and low free/ acyl carnitine ratio (0.66)
4. GCMS of urine: shows no specific abnormality
5. Plasma amino acids: shows mildly elevated glycine (717 umol/L) and alanine (834 umol/L)
6. CSF amino acids: shows mildly elevated glycine (55 umol/L)
7. CSF pipecolic acid: Normal
8. Urine AASA: Normal

The above results do not indicate any specific Inborn Error of Metabolism. Elevation of glycine can be seen due to anti- convulsants. Low free carnitine can be corrected by carnitine supplementation.

I would like to suggest the following:

1. MRI with MRS
2. Genetic studies

If you wish I can see this child during my next visit to Coimbatore- 13-15 February, 2023

Thanking you

Dr. Anil Jalan
MD DCH MCPS

BIOCHEMICAL GENETIC REPORT